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# Yenepoya (Deemed University)

## Project – High Level Design

## on

## Cloud Cost Intelligence Platform for Finance

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**HLD - Cloud Cost Intelligence Platform for Finance**

# 1. Introduction

This project is a Python-based data analysis system designed for the cloud finance domain. It enables structured analysis of cloud billing data using Pandas-based aggregation and Matplotlib visualisation, requiring no external APIs or internet connectivity. It accepts a cloud billing CSV export (cloud_billing_1year.csv) containing DATE, SERVICE TYPE, USAGE, and COST columns, and produces cost breakdowns by service and by month, a pie chart showing service cost distribution, and a bar chart showing the monthly spending trend across the full year 2025

## 1.1. Scope of the Document

This document covers the high-level architecture of the Cloud Cost Intelligence Pipeline, including: component interactions and data flow through the four-layer notebook pipeline; data cleaning and transformation logic overview; cost aggregation and groupby design; visualisation output design (pie chart and bar chart); offline deployment and runtime behaviour within Jupyter Notebook; and nonfunctional requirements at a high level.

## 1.2. Intended Audience

This document is intended for the following audiences:

• Developers building or extending the billing analysis pipeline

• Architects reviewing the data pipeline and component design decisions

• Industry Mentors from IBM evaluating the technical approach

• University Faculty assessing project completeness and analytical depth

• Finance teams or cloud operations personnel interpreting the cost reports

• Project Managers tracking scope and deliverables

### 1.3. System Overview

The solution is a lightweight offline data analysis application built with Python. It reads a flat CSV billing export, applies cleaning and transformation logic, computes cost aggregations using Pandas groupby operations, and renders Matplotlib charts inline within a Jupyter Notebook environment. The dataset processed contained 1,095 billing rows (3 services × 365 days) covering January 2025 to December 2025, with a total analysed cloud spend of $120,350

| Attribute | Detail |
|---|---|
| **System Type** | Offline data analysis pipeline — Jupyter Notebook |
| **Connectivity** | Fully offline — no internet or external API required |
| **Language** | Python 3.x |
| **Libraries** | Pandas, Matplotlib (standard data science stack) |
| **Storage** | In-memory Pandas DataFrames; local file I/O for input CSV |
| **Dataset** | cloud_billing_1year.csv — 1,095 rows × 4 columns (DATE, SERVICE TYPE, USAGE, COST) |
| **Analysis Period** | January 2025 – December 2025 (Full Calendar Year) |
| **Total Spend Analysed** | $120,350.00 USD |

*Figure 1: High Level System Architecture — Four-Layer Jupyter Pipeline (Ingest → Clean → Aggregate → Visualise)*

# 2. System Design

## 2.1. Application Design

The application follows these core design principles:

- Separation of Concerns — each pipeline stage (ingestion, cleaning, transformation, aggregation, visualisation) has a single well-defined responsibility executed in a distinct Jupyter cell block
- Offline-First Design — no external API calls; all data is loaded from a local CSV file, ensuring full functionality without internet connectivity
- Simplicity Over Complexity — uses only Pandas and Matplotlib; no third-party BI frameworks or database dependencies
- Stateful Session — intermediate DataFrames (data, service_cost, monthly_cost) persist in kernel memory across cells, enabling downstream cells to reference upstream outputs without re-computation
- Flat Schema Assumption — the pipeline assumes the four required columns (DATE, SERVICE TYPE, USAGE, COST) are present; column name stripping (str.strip()) adds resilience against whitespace-padded headers
- Notebook-Native UX — designed for sequential Jupyter cell execution with charts rendered inline via plt.show()

## 2.2. Process Flow

*Figure 2: Overall Process Flow — Sequential Cell Execution In [10] through In [35]*

| Step | Notebook Cell | Action Performed |
|---|---|---|
| 1 | In [10] | Import pandas as pd and matplotlib.pyplot as plt |
| 2 | In [25] | Read cloud_billing_1year.csv using pd.read_csv() — returned 1,095 rows × 4 columns |
| 3 | In [26] | Stripped whitespace from all column names using data.columns.str.strip() |
| 4 | In [27] | Removed rows with missing values using data.dropna() |
| 5 | In [28] | Parsed DATE column to datetime64 using pd.to_datetime(format='%d-%m-%Y') |
| 6 | In [29] | Derived Month integer column (1–12) from DATE using .dt.month — DataFrame expanded to 5 columns |
| 7 | In [31–32] | Computed service_cost via groupby('SERVICE TYPE')['COST'].sum() → EC2: $77,800 \| LAMBDA: $23,340 \| S3: $19,210 |
| 8 | In [33] | Computed monthly_cost via groupby('Month')['COST'].sum() → 12-row Series, Jan $3,797.50 to Dec $16,585.00 |
| 9 | In [34] | Rendered pie chart 'Cost by Service' with autopct='%1.1f%%' — EC2 64.6%, LAMBDA 19.4%, S3 16.0% |
| 10 | In [35] | Rendered bar chart 'Monthly Cost' — Month on x-axis, Cost (USD) on y-axis, consistent upward trend Jan–Dec |

## 2.3. Information Flow

*Figure 3: Information Flow Diagram — Unidirectional: CSV File → DataFrame → Aggregation → Chart*

| Source / Input | | Processor | | Output |
|---|---|---|---|---|
| cloud_billing_1year.csv | → | pd.read_csv() | → | Raw DataFrame 1,095 × 4 |
| Raw DataFrame | → | dropna(), str.strip(), to_datetime() | → | Clean DataFrame 1,095 × 5 |
| Clean DataFrame | → | groupby('SERVICE TYPE').sum() | → | service_cost — 3-row Series |
| Clean DataFrame | → | groupby('Month').sum() | → | monthly_cost — 12-row Series |
| service_cost | → | plt.pie() + autopct | → | Pie Chart: Cost by Service |
| monthly_cost | → | plt.bar() + axis labels | → | Bar Chart: Monthly Cost Trend |

## 2.4. Components Design

The system is composed of six core pipeline components:

DataLoader — reads cloud_billing_1year.csv using pd.read_csv() and returns a raw 1,095 × 4 DataFrame with columns DATE, SERVICE TYPE, USAGE, COST.

DataCleaner — applies str.strip() to column names to sanitise whitespace, then dropna() to remove incomplete rows, ensuring aggregations are computed only on valid records.

Transformer — converts DATE to datetime64 using pd.to_datetime() and derives a Month integer column (1–12) using .dt.month, expanding the DataFrame to 5 columns.

CostAggregator — executes two independent groupby().sum() operations to produce service_cost (3 rows: EC2, LAMBDA, S3) and monthly_cost (12 rows: Jan–Dec).

ChartEngine — renders a pie chart (service cost share with percentage labels) and a bar chart (monthly spend trend) using Matplotlib's Pyplot interface.

Control Loop (Notebook) — the Jupyter Notebook cell execution sequence orchestrates the full pipeline: Load → Clean → Transform → Aggregate → Visualise.

## 2.5. Key Design Considerations

01. Column Name Sanitisation — data.columns.str.strip() is applied immediately after loading to remove leading/trailing whitespace from column names. The column 'SERVICE TYPE' contains a space internally, so it must be referenced exactly as written after stripping outer whitespace; failure to strip causes KeyError on groupby.

02. In-Memory Storage — all DataFrames (data, service_cost, monthly_cost) are held in-memory within the Jupyter kernel session. This eliminates I/O latency but means data is lost on kernel restart; a full re-run is required after any restart.

03. Date Format Specification — pd.to_datetime() is called with format='%d-%m-%Y' to match the DD-MM-YYYY string format in the source CSV. Without this explicit format, Pandas may misparse dates (e.g., treating day as month).

04. Duplicate Groupby Execution — in the submitted notebook, service_cost was computed twice (In [31] and In [32]). Results are identical; this is a redundant cell that should be removed in the final clean submission.

05. Fallback Handling — dropna() acts as the fallback quality gate. Any row with a missing COST, DATE, or SERVICE TYPE is silently dropped before aggregation, preventing NaN propagation into the final cost totals.

## 2.6 API Catalogue

The following internal Pandas and Matplotlib function interfaces define the analytical API of the pipeline:

*Figure 4: Internal Function Interface — Deployment View*

| Function | Library | Purpose in This Project |
|---|---|---|
| read_csv(path) | pandas | Load billing CSV into raw DataFrame |
| columns.str.strip() | pandas | Sanitise column name whitespace |
| dropna() | pandas | Remove rows with any missing values |
| to_datetime(col, format=) | pandas | Parse DATE strings to datetime64 using DD-MM-YYYY format |
| .dt.month | pandas DatetimeIndex | Extract month integer (1–12) from datetime column |
| groupby(col)[col].sum() | pandas | Aggregate total cost by service or by month |
| plot(kind='pie', autopct=) | pandas / matplotlib | Render pie chart with percentage labels per slice |
| plot(kind='bar') | pandas / matplotlib | Render bar chart of monthly cost with x/y axis labels |
| plt.title() / xlabel() / ylabel() | matplotlib.pyplot | Add chart title and axis labels |
| plt.show() | matplotlib.pyplot | Render and display chart inline in Jupyter Notebook |

# 3. Data Design

## 3.1 Data Model

The billing dataset follows a flat daily-granularity model. Each row represents one cloud service's billing record for a single day. With 3 services billed every day for 365 days, the total expected row count is 1,095, which matches the loaded dataset exactly. After cleaning and transformation, the canonical DataFrame schema is:

| Column | Original Type | Processed Type | Description |
|---|---|---|---|
| DATE | object (string) | datetime64[ns] | Billing date parsed from DD-MM-YYYY format |
| SERVICE TYPE | object | object (stripped) | Cloud service name: EC2, S3, or LAMBDA |
| USAGE | int64 | int64 | Usage quantity (hours / requests / GB depending on service) |
| COST | float64 | float64 | Billed cost in USD — primary aggregation target |
| Month | — (derived) | int64 | Integer 1–12 extracted from DATE via .dt.month |

## 3.2 Data Access Mechanism

All data access is file-based and local. The DataLoader reads the input CSV from the Jupyter working directory using pandas.read_csv(). Processed DataFrames are held entirely in-memory (RAM) for the duration of the active kernel session. Chart outputs are displayed inline using plt.show(). No database, remote storage, or network endpoint is accessed at any stage of the pipeline.

## 3.3 Data Retention Policies

The input CSV (cloud_billing_1year.csv) is read-only and is never modified by the pipeline. Intermediate DataFrames exist only within the active Jupyter kernel and are not automatically persisted to disk. In the submitted notebook, charts are rendered inline via plt.show() without explicit file export. For production deployment, plt.savefig() and DataFrame.to_csv() should be used to save outputs to a versioned output directory (e.g., /outputs/2025-Q4/).

## 3.4 Data Migration

The pipeline processes exported billing snapshots and does not maintain a persistent database, so formal data migration is not applicable. Adapting the pipeline to a different cloud provider's billing format (e.g., GCP Billing Export or Azure Cost Management CSV) requires: (1) remapping column names to the canonical schema (DATE, SERVICE TYPE, USAGE, COST), and (2) adjusting the date format string in pd.to_datetime(). The aggregation and visualisation layers require no changes.

# 4. Interfaces

The system interfaces exclusively through local file I/O and the Jupyter Notebook interactive environment:

| Interface | Type | Format | Description |
|---|---|---|---|
| **Billing Input** | **File Read** | **CSV** | **cloud_billing_1year.csv — 1,095 rows, 4 columns** |
| **Pie Chart Output** | **Inline Display** | **Matplotlib** | **EC2 64.6% \| LAMBDA 19.4% \| S3 16.0% — rendered in In [34]** |
| **Bar Chart Output** | **Inline Display** | **Matplotlib** | **Monthly spend Jan–Dec 2025 — rendered in In [35]** |
| **User Interaction** | **Jupyter Notebook** | **Cell execution** | **Sequential cell execution — In [10] through In [35]** |

# 5. State and Session Management

The platform is stateless by design. All processing state resides in the Jupyter kernel's in-memory namespace. Three named objects persisted across cells during the analysis session: data (the primary cleaned DataFrame, 1,095 × 5), service_cost (the 3-row aggregated cost Series), and monthly_cost (the 12-row monthly cost Series). Cells must be executed in sequence (In [10] → In [35]). Executing a visualisation cell before its upstream aggregation cell would raise a NameError. A full kernel restart followed by 'Run All Cells' is the recommended execution pattern for reproducibility.

# 6. Caching

No explicit caching layer is implemented in the submitted notebook. The following implicit caching behaviours were present during the session:

- In-memory DataFrame caching: Once loaded in In [25], the data DataFrame remained in kernel memory across all subsequent cells without re-reading the source CSV file.
- Duplicate groupby execution: service_cost was computed in both In [31] and In [32]. Pandas executed the groupby operation twice; results were identical. This is an inefficiency to be resolved in the final clean submission by removing the duplicate cell.
- Figure memory: Matplotlib figure objects for both charts were held in memory until kernel shutdown. For extended sessions producing many charts, plt.close() should be called after plt.show() in each chart cell to release figure memory.

# 7. Non-Functional Requirements

## 7.1. Security Aspects

- Data Locality — the billing CSV was processed entirely on the local machine; no data was transmitted to any external endpoint or cloud service during the analysis.
- No Credentials Required — the pipeline required no cloud API keys, IAM roles, or authentication tokens; billing data was used solely as a local static file input.
- Sensitive Financial Data — the dataset contains actual cost figures ($120,350 total annual spend); the CSV and notebook output should be stored with restricted file permissions and excluded from public version control repositories.
- No Web Exposure — all chart outputs were rendered inline within Jupyter; no HTTP server or web dashboard was created, eliminating web-layer attack surfaces.

### 7.2. Performance Aspects

| Metric | Observed / Target | Notes |
|---|---|---|
| File load — 1,095 rows | < 1 second | Standard read_csv() on a small dataset; negligible I/O time |
| dropna() + to_datetime() | < 1 second | Vectorised Pandas operations; no row-level Python loops |
| groupby().sum() × 2 | < 1 second | 3-group and 12-group aggregations on 1,095 rows are trivial |
| Pie chart render (In [34]) | 2–4 seconds | Matplotlib render time; autopct label computation adds minor overhead |
| Bar chart render (In [35]) | 1–3 seconds | 12-bar chart; standard Matplotlib render time |
| Total pipeline runtime | < 10 seconds | Full notebook run In [10] to In [35] on standard hardware |

## 8. References

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6. University Project Guidelines 2026